In-Context Learning: Chain-of-Thought vs. Tree-of-Thoughts Prompting on Multi-Step Reasoning

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***Abstract*—We present an empirical comparison of four In-Context Learning (ICL) prompting strategies on the GSM8K grade-school mathematics benchmark—Zero-Shot Chain-of-Thought (CoT), Few-Shot CoT (8-shot), Tree-of-Thoughts (ToT) with BFS at two search depths, and Self-Consistency voting—using Llama 3.1 8B via the Groq API without fine-tuning. Zero-Shot CoT achieves 95.0%, Self-Consistency 92.0%, and ToT (BFS-2, depth=6) and Few-Shot CoT each 85.0% on the same 100-problem seed-42 sample. A single answer-extraction format instruction raises Zero-Shot CoT by 7 pp (88.0% $\rightarrow$ 95.0%) and reorders the strategies—a parsing confound, not a reasoning gain—placing it alone on the accuracy–cost Pareto frontier at the 8B scale. A depth ablation yields only $+4.0$ pp at $2.1\times$ cost, indicating diminishing returns when the evaluator is the same 8B backbone. A cross-model replication on Claude Haiku 4.5 with full evaluation identity preserved collapses the inter-strategy gap from 14 to 2 pp, revealing a *ceiling effect*: deliberation- and ensemble-style strategies pay off only when the base model has headroom to lose and identifiable wrong branches to prune. A pairwise Cohen's κ analysis confirms this—ToT depths on Llama reach near-zero agreement (κ=0.011) versus perfect agreement on Haiku (κ=1.000).**

***Index Terms*—Chain-of-thought, GSM8K, in-context learning, large language models, multi-step reasoning, self-consistency, tree-of-thoughts**

I. Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities in natural language tasks through In-Context Learning (ICL), where the model adapts to new tasks from demonstrations provided in the prompt without updating model weights [1]. This ability to generalise from a handful of examples has made ICL the dominant paradigm for practical LLM deployment, motivating active research into how the structure and content of prompts influence multi-step reasoning quality. Among ICL techniques, Chain-of-Thought (CoT) prompting has emerged as a particularly effective approach, guiding models to produce intermediate reasoning steps before arriving at a final answer [2]. The zero-shot variant [3] further simplified this by showing that a single trigger phrase suffices for instruction-tuned models to produce step-by-step reasoning, suggesting that structured reasoning has been internalised during pre-training.

Tree-of-Thoughts (ToT) extends this paradigm by exploring multiple reasoning branches simultaneously and pruning low-quality paths through a separate evaluation step [4]. While ToT has shown impressive gains over CoT on GPT-4, its effectiveness on smaller, open-weight models remains unexplored. A central open question is whether the multi-branch search ToT provides translates to accuracy gains when the evaluator model is the same small backbone—or whether evaluator weakness undermines the pruning mechanism entirely.

This paper presents a controlled empirical comparison of four prompting strategies (Zero-Shot CoT, Few-Shot CoT (8-shot), ToT with BFS, and Self-Consistency) on the GSM8K benchmark [5], using Llama 3.1 8B as a common backbone, complemented by a ToT depth ablation and a cross-model replication on Claude Haiku 4.5 (a frontier-tier model) under exact sample, prompt, and evaluation identity. Our three findings are summarised next as contributions.

a) Contributions: This paper makes three empirical contributions, all derived from a single controlled comparison rather than from new model training or larger experiments:

- **A scale-dependent ceiling effect.** Strategy ordering is not an intrinsic property of the prompting method but a function of backbone capability: the inter-strategy accuracy spread collapses from 14 pp on Llama 3.1 8B to 2 pp on Claude Haiku 4.5, and the *simplest* strategy on the strong backbone (Zero-Shot CoT, 97.0%) outperforms the *most elaborate* strategy on the weak one (Self-Consistency, 92.0%). A pairwise Cohen's κ analysis confirms this independently (Sections V-F–V-G).
- **Search depth is not the ToT bottleneck; evaluator discrimination is.** Doubling ToT search depth (3$\rightarrow$6) buys only $+4.0$ pp at $2.1\times$ inference cost on Llama and $+0.0$ pp on Haiku, isolating the weak *sure/maybe/impossible* self-evaluator—not insufficient depth—as the factor that prevents deliberate search from helping at the 8B scale (Section V-B).
- **Answer-extraction format is a confound in CoT benchmarking.** A single output-format instruction raises Zero-Shot CoT by $+7$ pp (88.0% $\rightarrow$ 95.0%) without

changing the reasoning process, showing that accuracy gaps attributed to a prompting *method* can partly reflect answer-extraction *format* and must be controlled when comparing strategies (Section IV).

II. Related Work

The foundational prompting strategy for multi-step reasoning is Chain-of-Thought (CoT), introduced by Wei et al. [2], who demonstrated that providing few-shot examples with step-by-step solutions substantially improves LLM performance on arithmetic and symbolic reasoning. Kojima et al. [3] subsequently showed that simply appending *"Let's think step by step"* to a query elicits CoT reasoning in a zero-shot setting, establishing a lightweight baseline that avoids hand-crafted demonstrations and anchors our Zero-Shot CoT configuration.

Building directly on CoT, Yao et al. [4] proposed Tree-of-Thoughts (ToT), which frames reasoning as a search problem over a tree of intermediate "thoughts". At each node, the model generates multiple candidate continuations, evaluates them with a separate scoring prompt (assigning *sure/maybe/impossible*), and retains only the most promising branches using BFS or DFS. Their experiments with GPT-4 showed substantial gains on tasks requiring deliberate planning; however, all results used models with 100B+ parameters, leaving the effectiveness of the same mechanism on smaller open-weight evaluators as an open question.

A complementary ensemble approach was proposed by Wang et al. [6], who showed that sampling diverse CoT paths at temperature > 0 and selecting the final answer by majority vote improves CoT accuracy by 10–20 pp without changing the prompt. Self-Consistency thus occupies a middle ground between the minimal-cost single-chain CoT and the expensive multi-branch ToT.

All three strategies are instances of the In-Context Learning paradigm of Brown et al. [1]—generalising to unseen tasks from a few in-context demonstrations without gradient updates—which makes them directly comparable under a fixed model and prompt template. Other structured-prompting variants (Automatic CoT [7], which clusters questions to auto-generate demonstrations, and Least-to-Most [8], which decomposes a problem into ordered subproblems) map out a broader design space; we focus on the four most widely adopted strategies for a controlled comparison.

III. Methodology

A. Zero-Shot Chain-of-Thought

The problem is presented with the trigger phrase *"Let's think step by step."* [3] plus the output-format instruction *"End your response with: The answer is X."* to prevent trailing-modifier parse errors (see Section IV, *Answer-extraction format fix*).

B. Few-Shot Chain-of-Thought (8-shot)

Eight worked examples from Wei et al. [2] are prepended to the query, each containing a full reasoning chain ending with *"The answer is X."* followed by the test problem.

C. Tree-of-Thoughts (BFS, branching=2, depth=6)

ToT (`src/run_tot_fixed.py`) explores a tree of intermediate "thoughts". At each depth level, two candidate next-step thoughts are generated at temperature 0.7; a separate evaluator prompt scores each as *sure*, *maybe*, or *impossible*, and only the former two are expanded. After BFS completion, a dedicated step extracts the final answer from the best leaf node. The configuration (branching factor 2, depth 6) uses $\sim$45 API calls per problem.

D. Self-Consistency (5 paths)

Few-Shot CoT is sampled 5 times per problem with temperature $= 0.7$. The final answer is selected by majority vote over the 5 sampled numeric answers [6].

E. Cross-Model Evaluation Protocol

To test whether strategy orderings generalise beyond the 8B scale, we re-run all five configurations on Claude Haiku 4.5 with identical 100-problem sample, prompts, parser, and gold-label correction, so any accuracy difference is attributable to the backbone alone. The three single-call strategies are batched through the Anthropic `messages.batches` API (50% discount); ToT runs synchronously because its BFS state cannot be flattened into batch items.

IV. Experimental Setup

A. Dataset

We use GSM8K [5], a benchmark of 8,792 grade-school math word problems requiring 2–8 chained reasoning steps. All five configurations (Zero-Shot CoT, Few-Shot CoT, ToT depth=3, ToT depth=6, and Self-Consistency) use the same 100-problem random sample of the test split (seed $= 42$). One item in our sample contains a gold-label annotation error (see the *GSM8K annotation error* paragraph below); we manually verified the correct value and use the corrected label in all evaluations, preserving the original sample sizes.

B. Model

All experiments use Llama 3.1 8B (`llama-3.1-8b-instant`) via the Groq inference API. No fine-tuning is applied; the model is used in its base instruction configuration.

C. Evaluation

The primary metric is *exact-match accuracy*: the predicted final number is compared against the gold answer after stripping commas and converting to float. Answers are extracted by a regular expression capturing the gold-format marker `####`, falling back to the last number otherwise. We also report the *parse failure rate* (responses yielding no numeric prediction). Llama 3.1 8B never spontaneously produced the `####` marker, so every response used the last-number fallback, which succeeded on all 490 evaluated responses (parse failure rate $= 0\%$).

D. Answer-Extraction Format Fix

An initial Zero-Shot CoT run sometimes ended with a trailing-modifier number rather than the substantive answer (e.g. "*36 hours in 4 weeks*" $\rightarrow$ parser captures `4` instead of `36`). Appending "*End your response with: The answer is X.*" to the prompt raised Zero-Shot CoT from **88.0%** to **95.0%** ($+7$ pp), concentrated in the hard bucket. All Zero-Shot CoT numbers use this format-aligned prompt; the other strategies already produce unambiguous final answers via their demonstrations or dedicated extraction prompts. Crucially, without this format alignment Zero-Shot CoT (88.0%) would appear merely tied with Few-Shot CoT and ToT (both 85.0%): the strategy ordering itself is altered by an extraction-format change that leaves the reasoning process untouched. Reported accuracy gaps between prompting strategies can therefore partly reflect answer-extraction format rather than reasoning ability, and the format must be held fixed—or reported—for a fair comparison (see Section VI).

E. GSM8K Annotation Error (Gold-Label Correction)

Manual inspection identified one annotation error in our sample (the *carnival* item): the official gold label \$2,280 omits one addend; the correct total is $430+750+300+700 =$ \$2,180. We adopt the corrected label for all reported accuracies; Zero-Shot CoT and ToT depth=6 (both produced 2,180) shift from incorrect to correct, consistent with the $\sim$1–2% annotation noise floor of GSM8K [5].

F. Infrastructure

Rate-limit constraints on the Groq free tier required an 8-second inter-call pause (CoT) and a 30-second inter-problem pause (ToT). All random seeds are fixed at 42 for reproducibility. All code and result files are publicly available.[1]

G. Secondary Model and Cross-Model Setup

The secondary backbone is Claude Haiku 4.5 (`claude-haiku-4-5-20251001`) via the Anthropic Python SDK, run on the same seed-42 sample, prompts, parser, and carnival gold-label correction as Llama, with batching as described in Section III-E (700 batched requests). Realized total cost: \$3.26 across all five strategies (\$0.52 batch + \$0.72 ToT-d3 + \$2.02 ToT-d6).

V. Results

A. Main Results

Table I summarises the accuracy of the four strategies.

Beyond accuracy, we report two efficiency metrics that have become standard in recent reasoning benchmarks [9], [10]: the average per-problem token budget $\bar{T}$ (a model-agnostic proxy for compute cost) and *tokens per correct answer* (TPC $= \bar{T}/$accuracy), which captures the practical cost of *useful* reasoning—a method achieving high accuracy but spending disproportionately many tokens may be impractical [4], [9]. Few-Shot CoT is the most token-efficient strategy (171 TPC), followed closely by Zero-Shot CoT (199); both ToT configurations spend 2–4$\times$ more tokens per correct answer (389 and 660), and Self-Consistency—despite the highest accuracy—is the most expensive at 802 TPC, consistent with its 5$\times$ sample budget.

Zero-Shot CoT achieves 95.0% (after the answer-extraction format fix, Section IV) and substantially outperforms Few-Shot CoT (85.0%), reversing the typical expectation that more examples improve performance. This suggests that Llama 3.1 8B has already internalised strong step-by-step reasoning during pre-training, and the 8-shot demonstrations introduce noise rather than useful signal for this sample.

ToT (depth=6) matches Few-Shot CoT at 85.0% but trails Zero-Shot CoT by 10.0 pp; Section V-F tests whether these orderings generalise to a stronger backbone.

B. Depth Ablation for Tree-of-Thoughts

To isolate the effect of search depth, we ran ToT again on the same 100-problem sample with `depth=3` (branching factor unchanged at 2), mirroring the short-horizon depth of the original ToT paper [4] against our deeper `depth=6` on identical problems and the same evaluator prompt. Table II reports the result and the accuracy–cost trade-off.

The $+4.0$ pp gain is small relative to the $2.1\times$ cost increase. Unlike GPT-4's 100B+ evaluator [4], our 8B *sure/maybe/impossible* scorer is weakly discriminative—most partial thoughts receive *sure* regardless of correctness, collapsing BFS toward a near-linear chain. GSM8K problems (2–8 steps) also fit comfortably within depth=3, so additional depth helps only a small tail of multi-step problems. The bottleneck at the 8B scale is therefore evaluator discrimination rather than search depth.

C. Self-Consistency: Agreement and Pass@k

We summarise two Self-Consistency diagnostics in text for brevity. *Vote agreement:* all five paths agreed unanimously on 65% of problems and reached a 4/5 or 3/5 majority on a further 30%, so 95% of problems admit a clean decision (88.2% average majority margin), confirming that 5 paths suffice for reliable consensus. *Pass@k* [11] (the fraction of problems where at least one of the first k paths is correct): Pass@1 = 87.0% matches greedy Few-Shot CoT (85.0%), so a single stochastic sample adds no consistent gain, whereas Pass@5 reaches 99.0%. This exposes a $+7.0$ pp *verification gap* over majority voting (92.0%): in 7 of 100 problems the correct answer is among the five sampled paths but loses the vote, indicating that a smarter aggregator—not more samples—is the main opportunity for further gains.

D. Accuracy Stratified by Problem Complexity

We adopt the Cobbe et al. [5] difficulty split: problems are bucketed by gold-solution operation count into *easy* (< 3), *medium* ($= 3$), and *hard* (> 3) groups (30/33/37 in our sample). Table III reports per-bucket accuracy.

Three results stand out. First, every strategy shows the expected easy-to-hard accuracy decay. Second, Few-Shot CoT

[1] https://github.com/OguzhanCKL/CENG467_In-Context_Learning

TABLE I

MAIN RESULTS ON GSM8K. LLAMA 3.1 8B, GROQ API. ACCURACIES USE THE GSM8K GOLD LABELS WITH ONE ITEM CORRECTED (SEE SECTION IV, *GSM8K annotation error*). $\bar{T}$ DENOTES THE AVERAGE RESPONSE LENGTH IN TOKENS (ONE TOKEN ≈ FOUR CHARACTERS), AGGREGATED ACROSS ALL API CALLS PER PROBLEM; TPC = TOKENS PER CORRECT ANSWER (LOWER IS BETTER).

Strategy	**Accuracy**	**N**	**API Calls**	$\bar{T}$	**TPC**	**Status**
Zero-Shot CoT	**95.0%**	100	$\sim$1	189	199	Complete
Few-Shot CoT (8-shot)	**85.0%**	100	$\sim$1	145	171	Complete
Tree-of-Thoughts (BFS-2, depth=3)	**81.0%**	100	$\sim$21	315	389	Complete (ablation)
Tree-of-Thoughts (BFS-2, depth=6)	**85.0%**	100	$\sim$45	561	660	Complete
Self-Consistency (5 paths)	**92.0%**	100	5	738	802	Complete

TABLE II

TOT DEPTH ABLATION ON GSM8K (LLAMA 3.1 8B, BFS-2, IDENTICAL EVALUATOR PROMPT). ACCURACIES USE THE CORRECTED GSM8K LABELS (SEE SECTION IV).

Configuration	**Accuracy**	**N**	**API calls per problem**	**Wall-clock per problem**
ToT BFS-2, depth=3	81.0%	100	$\sim$21	$\sim$3 min
ToT BFS-2, depth=6	85.0%	100	$\sim$45	$\sim$6 min
Δ (depth=6 − depth=3)	**+4.0 pp**	–	**+114%**	**+100%**

loses the most on hard problems (-16.2 pp vs. Zero-Shot CoT), since its short 2–3-step demonstrations do not transfer well to longer reasoning chains. Third, ToT (depth=3) collapses on medium problems (75.8%) because 3-operation problems exactly exhaust its depth budget, leaving no slack for BFS recovery after an initial error. After the format fix, Zero-Shot CoT ties Self-Consistency on easy and medium buckets and *exceeds* it on hard (94.6% vs. 86.5%, $+8.1$ pp)—Zero-Shot CoT with disciplined answer extraction is the strongest strategy at the 8B scale across all difficulty buckets.

E. Solution Length Ratio: Reasoning Verbosity and Overthinking

Token cost ($\bar{T}$, TPC) does not reveal whether compute is well-spent. We quantify reasoning verbosity via the **Solution Length Ratio** SLR $=$ (model reasoning steps)/(gold solution steps), where model steps count "`= NUMBER`" patterns and gold steps are the GSM8K «`...`» annotations (SLR>1 indicates overthinking [12]; average gold length is 3.3 steps). Self-Consistency is the most compact strategy ($1.27\times$), even below Few-Shot CoT ($1.44\times$) and Zero-Shot CoT ($1.60\times$), because averaging across five paths filters out the longest chains. ToT (depth=6), by contrast, inflates solution length to $3.30\times$ (depth=3: $2.72\times$) while only matching Few-Shot CoT in accuracy—a clean instance of the *overthinking tax* [12]: BFS expansions add depth without semantic gain because the 8B evaluator cannot prune unproductive branches.

F. Cross-Model Comparison: Llama 3.1 8B vs. Haiku 4.5

To test whether strategy orderings reflect intrinsic prompting properties or 8B-scale limitations, we re-ran every strategy on Claude Haiku 4.5 with full evaluation identity preserved (Sections III-E–IV-G).

a) Headline accuracy: Table IV reports overall accuracy on the 100-problem seed-42 sample under the gold-corrected labels for both models.

Every strategy improves on the stronger backbone, as expected; the more interesting observation is that the *gaps between strategies collapse*: the five strategies span 14 pp on Llama (81.0% to 95.0%) but only 2 pp on Haiku (97.0% to 99.0%). The simplest strategy on Haiku (Zero-Shot CoT, 97.0%) already outperforms the most elaborate strategy on Llama (Self-Consistency, 92.0%) by 5 pp—at GSM8K's difficulty, raw backbone capability dominates prompting-strategy choice once the backbone is strong enough.

b) Stratified view: The Haiku columns of Table III report the same Cobbe difficulty buckets alongside the Llama numbers. Four of five strategies achieve a perfect 63/63 on Easy and Medium; Haiku's entire residual error budget lives in the Hard bucket. Only Self-Consistency recovers one additional Hard answer (36/37 vs. 35/37), mirroring its 8B advantage—Hard accuracy is the sole discriminating metric once the backbone nears ceiling.

c) Cross-model depth ablation: ceiling effect: Table V shows the sharpest single contrast: depth=3→6 gains $+4.0$ pp on Llama but 0.0 pp on Haiku at $2.8\times$ the token cost. On Haiku neither depth beats Zero-Shot CoT on Hard (both 94.6%); ToT's depth parameter buys accuracy only when the base model has headroom to lose and the evaluator has branches to discriminate.

d) Self-Consistency verifier saturation: On Llama the verifier gap is $+7.0$ pp (Pass@5 = 99%, majority-vote = 92%): the correct answer is present in the five paths but discarded by majority voting. On Haiku the gap is 0.0 pp (Pass@5 = majority-vote = 98%): on the two remaining errors, *none* of the five paths produces the correct answer. Path diversity has collapsed (98/100 problems show unanimous 5/5 voting), confirming that Self-Consistency's headroom is a model-scale artefact, not an intrinsic strategy benefit.

G. Strategy Agreement: Pairwise Cohen's κ Across Backbones

Accuracy alone does not reveal whether two strategies are correct on the *same* problems. We therefore compute all pairwise **Cohen's** κ [13] over the same 100-problem gold-corrected sample, treating each strategy as a binary classifier; $\kappa = 0$ is chance, $\kappa = 1$ is perfect agreement (Landis–Koch ranges [14]; see also Artstein and Poesio [15] on agreement coefficients for language data).

TABLE III
ACCURACY STRATIFIED BY PROBLEM COMPLEXITY (COBBE ET AL. [5] SPLIT, GOLD-CORRECTED; N=100, BUCKETS 30/33/37) ON BOTH BACKBONES. LLAMA COLUMNS ARE DISCUSSED IN THIS SECTION; HAIKU COLUMNS IN SECTION V-F. ALL VALUES ARE PERCENTAGES; EASY/MEDIUM/HARD = <3 / =3 / >3 GOLD OPERATIONS.

	Llama 3.1 8B				Haiku 4.5			
Strategy	**Easy**	**Med**	**Hard**	**Overall**	**Easy**	**Med**	**Hard**	**Overall**
Zero-Shot CoT	**96.7**	**93.9**	**94.6**	**95.0**	**100**	97.0	94.6	97.0
Few-Shot CoT (8-shot)	90.0	87.9	78.4	85.0	**100**	**100**	94.6	98.0
ToT (BFS-2, depth=3)	86.7	75.8	81.1	81.0	**100**	**100**	94.6	98.0
ToT (BFS-2, depth=6)	90.0	84.8	81.1	85.0	**100**	**100**	94.6	98.0
Self-Consistency (5 paths)	96.7	93.9	86.5	92.0	**100**	**100**	**97.3**	**99.0**

TABLE IV
CROSS-MODEL ACCURACY (GOLD-CORRECTED) ON THE SAME 100-PROBLEM GSM8K SAMPLE. "COST" IS THE REALIZED USD COST FOR THE FULL 100-PROBLEM RUN ON HAIKU 4.5; ZS/FS/SC SHARE A SINGLE 700-REQUEST BATCH WITH THE 50% BATCH DISCOUNT.

Strategy	**Llama 3.1 8B**	**Haiku 4.5**	**Δ**	**Haiku cost**
Zero-Shot CoT	95.0%	**97.0%**	+2.0	
Few-Shot CoT (8-shot)	85.0%	**98.0%**	+13.0	\$0.52 (batch)
Self-Consistency (5 paths)	92.0%	**99.0%**	+7.0	
ToT (BFS-2, depth=3)	81.0%	**98.0%**	+17.0	\$0.72 (sync)
ToT (BFS-2, depth=6)	85.0%	**98.0%**	+13.0	\$2.02 (sync)
			Total realized cost on Haiku 4.5:	**\$3.26**

TABLE V
CROSS-MODEL TOT DEPTH ABLATION (Δ = GOLD-CORRECTED ACCURACY).

Backbone	**depth=3**	**depth=6**	**Δ**
Llama 3.1 8B	81.0%	85.0%	+4.0 pp
Haiku 4.5	98.0%	98.0%	0.0 pp
Haiku token cost	depth=6 is 3.1× input and 2.4× output of depth=3		

TABLE VI
PAIRWISE COHEN'S κ BETWEEN THE FIVE STRATEGIES (100-PROBLEM SEED-42 SAMPLE, GOLD-CORRECTED). **LOWER TRIANGLE:** LLAMA 3.1 8B; **UPPER TRIANGLE:** HAIKU 4.5.

	ZS	**FS**	**SC**	**d=3**	**d=6**
ZS-CoT	–	0.385	0.492	**0.795**	**0.795**
FS-CoT	0.243	–	0.662	0.490	0.490
SC (5)	0.262	**0.563**	–	0.662	0.662
ToT d=3	0.095	0.435	0.207	–	**1.000**
ToT d=6	0.027	0.216	0.078	**0.011**	–

The two backbones invert. On Llama (lower triangle), the two ToT depths reach near-zero pairwise agreement ($\kappa = 0.011$) despite sharing the same BFS-2 algorithm and evaluator prompt—BFS exploration at the 8B scale is stochastic rather than deliberate—while Zero-Shot CoT shows only fair-to-slight agreement with every other strategy ($\kappa \in [0.027, 0.262]$), the genuine complementarity that drives the +7.0 pp Pass@5 headroom (Section V-C). On Haiku (upper triangle) the picture reverses: the two ToT depths reach $\kappa = 1.000$ (perfect problem-by-problem concordance) and both agree with Zero-Shot CoT at $\kappa = 0.795$, so deliberate-search machinery collapses into single-chain behaviour. The overall range collapses from [0.011, 0.563] on Llama to [0.385, 1.000] on Haiku: *strategy independence itself decays with model scale*.

H. Ablation Study

Two secondary Few-Shot CoT ablations (N=50, seed=42) are summarised in text. *Temperature:* greedy decoding (88.0%) outperforms stochastic sampling at T=0.7 (86.0%) by 2 pp, a marginal but consistent advantage for deterministic decoding on arithmetic tasks. *Prompt sensitivity:* accuracy across three 8-shot example sets spans 6 pp (82.0%–88.0%)—Set-1 underperforms (82.0%) while the canonical Wei et al. examples and Set-2 both reach 88.0%—indicating moderate sensitivity to example choice.

VI. DISCUSSION

Zero-shot CoT wins; Self-Consistency no longer dominates.

The 10-pp gap between Zero-Shot and Few-Shot CoT (Table I) indicates that Llama 3.1 8B already internalised step-by-step reasoning during pre-training, so the 8-shot demonstrations inject systematic noise rather than signal—a prompt-format mismatch that is actively harmful, not merely neutral. Once the answer-extraction fix lifts Zero-Shot CoT to 95.0%, Self-Consistency (at 5× the cost) is strictly Pareto-dominated: its per-path Pass@1 (87.0%) sits below the greedy chain.

Accuracy–cost trade-off.

On a log-scale cost axis (average API calls per problem), Zero-Shot CoT (∼1 call, 95.0%) lies alone on the Pareto frontier; every other configuration is dominated—Self-Consistency costs 5× for −3.0 pp, Few-Shot CoT matches the cost at −10.0 pp, and ToT costs 21–45× for −10 to −14 pp (see

$\bar{T}$/TPC in Table I). Where accuracy is paramount and cost secondary, Self-Consistency is the practical choice; otherwise Zero-Shot CoT offers the best accuracy-per-call ratio.

ToT and the ceiling effect.

ToT (depth=6) matches but never beats the CoT baselines on either backbone, so the absence of a ToT advantage is not an artefact of an undersized evaluator. Two factors explain this on Llama: GSM8K problems resolve in 3–5 steps that CoT already handles, and the *sure/maybe/impossible* evaluator (the same 8B model) is too weak to discriminate, collapsing BFS toward a near-linear chain (the depth ablation buys only $+4.0$ pp at $2.1\times$ cost). The Haiku replication sharpens this into a *ceiling effect*: the inter-strategy spread collapses from 14 pp to 2 pp, the depth=3→6 gain falls to 0.0 pp, and the Self-Consistency verifier gap shrinks from $+7.0$ to 0.0 pp because no sampled path recovers the problems voting gets wrong. The pairwise κ analysis confirms this independently (Table VI): Llama's ToT depths agree at only $\kappa = 0.011$ (stochastic exploration) versus $\kappa = 1.000$ on Haiku (deterministic ceiling). Deliberation- and ensemble-style strategies thus buy accuracy only when the base model has headroom to lose and identifiable wrong branches to prune.

Answer extraction is a confound, not a detail.

Before the format fix, Zero-Shot CoT (88.0%) was statistically indistinguishable from Few-Shot CoT and ToT (depth=6) at 85.0%, and a reader would reasonably conclude that the three strategies perform alike at the 8B scale. After appending a single output-format instruction—which changes nothing about how the model reasons—Zero-Shot CoT becomes the clear leader at 95.0%, and the gain concentrates in the hard bucket where trailing-modifier numbers are most common. The methodological implication is that, especially for smaller models, benchmark rankings can be sensitive to answer-extraction format rather than to the reasoning method under study; comparisons across prompting strategies should hold the extraction format fixed or report it explicitly, lest a parsing artefact masquerade as a method effect.

Error analysis and limitations.

Across all strategies and both backbones, the dominant failure mode is arithmetic: the model builds the correct reasoning structure but miscomputes one step, and the error propagates—on Haiku every residual error is of this kind, confirming that arithmetic brittleness persists even at frontier scale, only attenuated. The main limitations are: (1) exact-match accuracy ignores reasoning quality; (2) the cross-model comparison covers only two families—adding Mistral, Qwen, or Gemma at the 7–8B scale would separate pre-training data quality from raw scale; and (3) GSM8K's short arithmetic horizon means these findings should not be extrapolated to long-horizon planning benchmarks where ToT reported larger gains [4].

VII. Conclusion

We evaluated Zero-Shot CoT, Few-Shot CoT, ToT (at two search depths), and Self-Consistency on GSM8K across Llama 3.1 8B and Claude Haiku 4.5. On Llama, Zero-Shot CoT with disciplined answer extraction (95.0%) is both the most accurate and the most cost-efficient strategy—a strict Pareto dominator—while ToT only matches Few-Shot CoT and its depth ablation yields diminishing returns ($+4.0$ pp at $2.1\times$ cost) due to weak 8B-evaluator discrimination rather than insufficient search depth. We further show that this format alignment is not a cosmetic detail but a confound in CoT benchmarking: the same $+7$ pp it contributes reorders the strategies, so reported method gaps can partly reflect answer-extraction format and must be controlled. Replicating all five strategies on the stronger backbone with full evaluation identity preserved exposes a *ceiling effect*: the inter-strategy gap collapses from 14 to 2 pp, the ToT depth gain vanishes, and the Self-Consistency verifier gap closes—deliberation- and ensemble-style strategies buy real accuracy only when the base model has headroom to lose and identifiable wrong branches to prune.

a) Future work: Replacing the categorical *sure/maybe/impossible* evaluator with a learned scorer is the most direct route to better ToT on 8B-class models, and the 7.0 pp Pass@5–vote verification gap on Llama suggests that an external verifier—rather than larger ensembles—is the main opportunity at fixed sample budget. Extending the comparison to additional open-weight families (Mistral, Qwen, Gemma at the 7–8B scale) would disentangle pre-training data quality from raw scale and test whether the ceiling effect generalises.

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